Lecture 9 (*March 9*)

Reading: Section 3.4
Viewing: Lecture 9

For vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n \in \mathbb{R}^m$, recall that the subspace V spanned by these vectors was explained in Definition 5. Specifically, vectors in the subspace $V \subset \mathbb{R}^m$ are arbitrary linear combinations:

$$\alpha_1 \boldsymbol{v}_1 + \cdots + \alpha_n \boldsymbol{v}_n \tag{73}$$

for various scalars $\alpha_1, \ldots, \alpha_n$. But what if one of the vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ was itself a linear combination of the other vectors? For example, we could have:

$$\boldsymbol{v}_i = \beta_1 \boldsymbol{v}_1 + \cdots + \beta_{i-1} \boldsymbol{v}_{i-1} + \beta_{i+1} \boldsymbol{v}_{i+1} + \cdots + \beta_n \boldsymbol{v}_n$$

for some $i \in \{1, \ldots, n\}$. Then any linear combination of the form (73) can be written as a linear combination of the vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{i-1}, \boldsymbol{v}_{i+1}, \ldots, \boldsymbol{v}_n$ only:

$$\alpha_1 \boldsymbol{v}_1 + \cdots + \alpha_{i-1} \boldsymbol{v}_{i-1} + \alpha_i \underbrace{(\beta_1 \boldsymbol{v}_1 + \cdots + \beta_{i-1} \boldsymbol{v}_{i-1} + \beta_{i+1} \boldsymbol{v}_{i+1} + \cdots + \beta_n \boldsymbol{v}_n)}_{\boldsymbol{v}_i} + \alpha_{i+1} \boldsymbol{v}_{i+1} + \cdots + \alpha_n \boldsymbol{v}_n =$$

$$= (\alpha_1 + \alpha_i \beta_1) \boldsymbol{v}_1 + \cdots + (\alpha_{i-1} + \alpha_i \beta_{i-1}) \boldsymbol{v}_{i-1} + (\alpha_{i+1} + \alpha_i \beta_{i+1}) \boldsymbol{v}_{i+1} + \cdots + (\alpha_n + \alpha_i \beta_n) \boldsymbol{v}_n$$

So we don't need the vector $\boldsymbol{v}_i$ to span the subspace $V \subset \mathbb{R}^m$: this subspace is in fact spanned by the slightly smaller collection of vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{i-1}, \boldsymbol{v}_{i+1}, \ldots, \boldsymbol{v}_n$.

Definition 8. *Vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n \in \mathbb{R}^m$ are called* ***(linearly) independent*** *if none of them is a linear combination of the others. Equivalently, this means that any linear combination is non-zero:*

$$\boxed{\alpha_1 \boldsymbol{v}_1 + \cdots + \alpha_n \boldsymbol{v}_n \neq 0} \tag{74}$$

except for the zero combination (namely the one where $\alpha_1 = \cdots = \alpha_n = 0$).

If the vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n \in \mathbb{R}^m$ are not independent, we call them **(linearly) dependent**, which means that there exists a linear combination:

$$\alpha_1 \boldsymbol{v}_1 + \cdots + \alpha_n \boldsymbol{v}_n = 0 \tag{75}$$

where not all the $\alpha_1, \ldots, \alpha_n$ are 0. For example, if $\alpha_i \neq 0$, then we can rewrite equation (75) as:

$$\boldsymbol{v}_i = -\frac{\alpha_1}{\alpha_i} \boldsymbol{v}_1 - \cdots - \frac{\alpha_{i-1}}{\alpha_i} \boldsymbol{v}_{i-1} - \frac{\alpha_{i+1}}{\alpha_i} \boldsymbol{v}_{i+1} - \cdots - \frac{\alpha_n}{\alpha_i} \boldsymbol{v}_n$$

We conclude that a collection of vectors is linear dependent precisely when one of them can be written as a linear combination of the others.

For example, the two vectors:

$$\begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 \\ -3 \\ -1 \end{bmatrix} \tag{76}$$

are linearly independent, because otherwise one of them would be a linear combination (i.e. a multiple) of the other, and this is visibly not true. Geometrically, we can see this because the two vectors above are not on the same line passing through the origin. However, the three vectors:

$$\begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 \\ -3 \\ -1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 4 \\ -1 \\ -1 \end{bmatrix} \tag{77}$$

are linearly dependent, because the following linear combination is 0:

$$1 \cdot \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 1 \\ -3 \\ -1 \end{bmatrix} - 1 \cdot \begin{bmatrix} 4 \\ -1 \\ -1 \end{bmatrix} = 0 \tag{78}$$

Alternatively, you could think of this as saying that one of the three vectors is a linear combination of the other two vectors, e.g.:

$$\begin{bmatrix} 4 \\ -1 \\ -1 \end{bmatrix} = 1 \cdot \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 1 \\ -3 \\ -1 \end{bmatrix} \tag{79}$$

Definition 9. *A **basis** of a vector space V is a collection of <u>independent</u> vectors which span V.*

For example, both the collection (76) and the collection (77) span the plane:

$$\Big\{(x, y, z) \text{ such that } 2x - 3y + 11z = 0\Big\} \subset \mathbb{R}^3 \tag{80}$$

But among these two collections of vectors, only (76) is a basis, because they are linearly independent. The collection of vectors (77) is not a basis, because each of the three vectors in (77) can be written as a linear combination of the other two (see (79)). So any of these three vectors is redundant in spanning the plane (80), and that is why we don't need all three of them to have a basis.

Remark. *A basis of any vector space is not unique. In fact, any collection of two of the three vectors in* (77) *form a basis of the subspace* (80)*.*

Definition 10. *The **dimension** of a vector space V is the number of vectors in a basis of V.*

Implicit in the Definition above is that all bases of a vector space (and there are infinitely many of them, see the Remark above) have the same number of vectors in them. The abstract notion of dimension introduced above precisely matches our usual, geometric, notion of dimension: a line has dimension 1 (it is spanned by a single vector) and a plane has dimension 2 (it is spanned by two vectors, for example the plane (80) with the basis (76)).

So, practically, how can we tell when a collection of vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ is linearly independent? And if they are not linearly independent, how can we remove some of these vectors, in such a way as to obtain a basis of the subspace they span? The way to answer both of these questions is to construct the $m \times n$ matrix whose columns are the vectors in question:

$$A = \left[\ \boldsymbol{v}_1 \ \middle|\ \ldots \ \middle|\ \boldsymbol{v}_n\ \right] \tag{81}$$

For any vector $\boldsymbol{w} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$, we have seen that:

$$A\boldsymbol{w} = \alpha_1 \boldsymbol{v}_1 + \cdots + \alpha_n \boldsymbol{v}_n \tag{82}$$

Therefore, the vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ are linearly independent if and only if $A\boldsymbol{w} = 0$ happens only when $\boldsymbol{w} = 0$, or in other words, if $N(A) = 0$. Conversely, the vectors are dependent precisely when there exists $\boldsymbol{w} \neq 0$ such that $A\boldsymbol{w} = 0$. So to find a linear combination (82) which is 0, one only needs to find a non-zero vector $\boldsymbol{w} \in N(A)$, and this can be done by putting A in reduced row echelon form.

More concretely, given a collection of vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$, how do we find a subset of them which are linearly independent and thus give a basis of the vector subspace they span? Simple: put the given vectors into a matrix A (see (81)), put A in row echelon form, identify the pivot columns, and look at the corresponding columns of A. Those will be a basis. For example, consider the matrix:

$$A = \begin{bmatrix} 2 & 1 & 4 \\ 5 & -3 & -1 \\ 1 & -1 & -1 \end{bmatrix} \quad \overset{\text{REF}}{\rightsquigarrow} \quad U = \begin{bmatrix} \boxed{2} & 1 & 4 \\ 0 & \boxed{-5.5} & -11 \\ 0 & 0 & 0 \end{bmatrix}$$

The pivot columns are the first and second ones, so the upshot is that a basis of $C(A)$ consists of the first two columns of A. This is precisely the basis we found in (76).

Fact 7. *The rank of the matrix A (i.e. the number of pivot columns) equals the dimension of $C(A)$.*

Above, we saw how to find a basis of the vector space spanned by a collection of vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$. But there are other ways to present vector spaces, namely by equations. For example consider:

$$V = \Big\{ 2x - 2y - 6z - 4t = 5x + y - 3z + 8t = -3x + 2y + 7z + 3t = 0 \Big\} \subset \mathbb{R}^4$$

Since V is cut out by three equations in four unknowns, it is tempting to believe that V is one-dimensional. But not so fast, since there may be redundancy in these equations! Let's find a basis of V systematically. The way to do so is to think of V as the nullspace of the coefficient matrix:

$$V = N(A) \quad \text{where} \quad A = \begin{bmatrix} 2 & -2 & -6 & -4 \\ 5 & 1 & -3 & 8 \\ -3 & 2 & 7 & 3 \end{bmatrix}$$

To describe this nullspace, put the matrix A in reduced row echelon form:

$$A = \begin{bmatrix} 2 & -2 & -6 & -4 \\ 5 & 1 & -3 & 8 \\ -3 & 2 & 7 & 3 \end{bmatrix} \quad \overset{\text{RREF}}{\rightsquigarrow} \quad R = \begin{bmatrix} \boxed{1} & 0 & -1 & 1 \\ 0 & \boxed{1} & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Since $N(A) = N(R)$, the nullspace of the matrix R in reduced row echelon form can be described in terms of pivot variables (x and y) and free variables (z and t):

$$R \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = 0 \quad \text{i.e.} \quad \begin{cases} x - z + t = 0 \\ y + 2z + 3t = 0 \end{cases}$$

Hence:

$$V = \left\{ \begin{bmatrix} z-t \\ -2z-3t \\ z \\ t \end{bmatrix} \text{ for any } z, t \right\}$$

To find basis vectors of V, just set one of the free variables (in this case, either z or t) equal to 1 and the other free variable equal to 0. We conclude that a basis of V is given by:

$$\begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -1 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$

Therefore, V is two-dimensional, hence a plane.

Let us summarize the two ways of thinking about a vector subspace $V \subset \mathbb{R}^m$: either as the column space of a matrix A (when you are given a set of vectors which spans V), or as the nullspace of a matrix A (when you are given equations satisfied by V). In either case, analyzing V (i.e. finding a basis and computing its dimension) is systematically done by putting the matrix A in (reduced) row echelon form.

Remark. *If A is a square $n \times n$ matrix, the rank always satisfies $r \leq n$. We can only have equality, i.e. $r = n$, if the columns of A are linearly independent. This is equivalent to the only solution of $A\boldsymbol{v} = 0$ being the zero vector $\boldsymbol{v} = 0$, so the nullspace of A just consists of the zero vector. Moreover:*

$$\boxed{r = n} \quad \textit{if and only if} \quad \boxed{A \textit{ is invertible}} \tag{83}$$

Indeed, everything we said in this Remark can be translated as saying that (for A a square matrix) the system $A\boldsymbol{v} = \boldsymbol{b}$ has a single solution $\boldsymbol{v}$ for any vector $\boldsymbol{b}$. This solution is precisely $\boldsymbol{v} = A^{-1}\boldsymbol{b}$.